

Name _____

Date _____

Period _____

Applying Exponent Laws*Question Bank 3.05*

Show your work. Circle your final answer.

For questions 1 through 4, rewrite each expression as an equivalent expression with a positive exponent.

1. $\left(\frac{3}{4}\right)^{-3}$

2. $\left(-\frac{7}{8}\right)^{-2}$

3. $\left(\frac{-4}{5}\right)^{-4}$

4. $\left(\frac{22}{7}\right)^{-7}$

Matching.

5. Match each expression on the left with the equivalent expression on the right by writing the letter.

a. $\left(-\frac{11}{10}\right)^{-3}$ A. $\left(-\frac{11}{10}\right)^3$ B. $\left(\frac{10}{11}\right)^3$ C. $\left(\frac{11}{10}\right)^3$ D. $\left(-\frac{10}{11}\right)^3$

b. $\left(\frac{11}{10}\right)^{-3}$

c. $\left(-\frac{10}{11}\right)^{-3}$

d. $\left(\frac{10}{11}\right)^{-3}$

For questions 6 through 9, generate an equivalent expression with one base and one positive exponent. Show the laws you use.

6. $\left(-\frac{14}{17}\right)^{-2} \cdot \left(-\frac{14}{17}\right)^{-4}$

7. $\left(\frac{5}{8}\right)^{-2} \div \left(\frac{5}{8}\right)^{-4}$

8. $\left[\left(\frac{2}{3}\right)^5 \cdot \left(\frac{2}{3}\right)^{-2}\right]^3$

9. $\left(\frac{2.5^5}{2.5^3}\right)^{-1}$

Equivalent or not.

10. Consider the expression $\frac{(\frac{1}{6})^3}{6^{-5}} \cdot 6^4$. For each expression below, decide whether it is equivalent, and justify with a law of exponents.

a. $\frac{6^{-3}}{6^{-5}} \cdot 6^4$

b. $6^{-8} \cdot 6^4$

c. $\left(\frac{1}{6}\right)^{-2} \cdot 6^4$

d. 6^6

11. Which is the value of $\left(-\frac{2}{3}\right)^{-3}$?

☐ A. $-\frac{27}{8}$

☐ B. $\frac{27}{8}$

☐ C. $-\frac{8}{27}$

☐ D. $\frac{8}{27}$

12. Find the value of n that makes each statement true.

a. $\left(\frac{3}{5}\right)^n = \frac{25}{9}$

b. $\left(\frac{1}{2}\right)^n = 8$